THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 10: **Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The length of the tangent from an external point P to a circle is 16 cm and P is 34 cm from the centre. The radius of the circle is: 1

(a) 50 cm **(b)** 18 cm **(c)** 30 cm **(d)** 17 cm

2. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 30 cm. The length of each tangent is 16 cm. The perimeter of the quadrilateral $PAOB$ is: 1

(a) 136 cm **(b)** 32 cm **(c)** 46 cm **(d)** 92 cm

3. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 40 cm. The length of each tangent is 9 cm. The area of the quadrilateral $PAOB$ (in cm^2) is: 1

(a) 360 cm² **(b)** 98 cm² **(c)** 720 cm² **(d)** 180 cm²

4. Two concentric circles have radii 52 cm and 20 cm. A chord of the larger circle touches the smaller circle. The length of the chord is: 1

(a) 96 cm **(b)** 72 cm **(c)** 48 cm **(d)** 64 cm

5. In $\triangle ABC$, right-angled at B, $AB = 11$ cm and $BC = 60$ cm. A circle is inscribed in the triangle. The radius of the inscribed circle is: 1

(a) 10 cm **(b)** 5 cm **(c)** 30 cm **(d)** 6 cm

6. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (8x + 3)$ cm and $QB = (5x + 18)$ cm. The length of QA is: 1

(a) 43 cm **(b)** 45 cm **(c)** 58 cm **(d)** 5 cm

7. PA and PB are tangents from an external point P to a circle, and $\angle PAB = 35^\circ$. The $\angle APB$ is: 1

(a) 110° **(b)** 55° **(c)** 70° **(d)** 35°

8. The length of a tangent from a point P to a circle is 12 cm and the radius is 5 cm. The distance of P from the centre is: 1

(a) 13 cm **(b)** 7 cm **(c)** 17 cm **(d)** $\sqrt{119}$ cm

9. The length of the tangent from an external point P to a circle is 21 cm and P is 35 cm from the centre. The radius of the circle is: 1

(a) 14 cm **(b)** 22 cm **(c)** 56 cm **(d)** 28 cm

10. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 28 cm. The length of each tangent is 21 cm. The perimeter of the quadrilateral $PAOB$ is: 1

(a) 98 cm **(b)** 133 cm **(c)** 49 cm **(d)** 42 cm

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. PA and PB are tangents from an external point P to a circle, and $\angle PAB = 30°$. Find the $\angle APB$. **2**

12. PA and PB are tangents from an external point P to a circle with centre O, touching it at A and B. $\angle APB = 36°$. Find the $\angle OAB$. **2**

13. A point P is 52 cm from the centre O of a circle of radius 48 cm. Find the length of the tangent drawn from P to the circle. **2**

14. PA and PB are tangents from an external point P to a circle, and $\angle PAB = 55°$. Find the $\angle APB$. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (9x + 4)$ cm and $QB = (5x + 20)$ cm. Find the length of QA. **3**

16. A quadrilateral $ABCD$ is drawn about a circle so that all four sides touch the circle. $AB = 18$ cm, $BC = 15$ cm and $CD = 12$ cm. Find the length of AD. **3**

17. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 30 cm. The length of each tangent is 16 cm. Find the area of the quadrilateral $PAOB$ (in cm^2). **3**

OR

Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (7x + 2)$ cm and $QB = (4x + 14)$ cm. Find the length of QA. **3**

18. A quadrilateral $ABCD$ is drawn about a circle so that all four sides touch the circle. $AB = 21$ cm, $BC = 17$ cm and $CD = 14$ cm. Find the length of AD. **3**

OR

Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 28 cm. The length of each tangent is 21 cm. Find the area of the quadrilateral $PAOB$ (in cm^2). **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Tangents PA and PB are drawn from a point P at a distance of 52 cm from the centre O of a circle of radius 20 cm. Find the length of each tangent, the perimeter and the area of the quadrilateral $PAOB$ (in this order). **5**

OR

Prove that if PA and PB are tangents from P to a circle with centre O, then $\angle APB + \angle AOB = 180°$. **5**

20. Tangents PA and PB are drawn from a point P at a distance of 53 cm from the centre O of a circle of radius 28 cm. Find the length of each tangent, the perimeter and the area of the quadrilateral $PAOB$ (in this order). **5**

OR

Prove that if PA and PB are tangents from P to a circle with centre O, then OP is the perpendicular bisector of AB. **5**

END OF QUESTION PAPER — Total: 40 Marks